Faculty of Engineering & Technology Electrical & Computer Engineering Department

ENCS3340

Comparative Study of Image Classification Using Decision Tree, Naive Bayes, and Feedforward Neural Networks

Project #2

Prepared by :

Student Name : Aya Abdullah **ID:** 1220782

Student Name : Lana Daramna **ID:**1220588

Instructor: Samah Alaydi

Section : 2

Date : 24-June-2025

Abstract

This project details the development and comparisons of three classification models Naive Bayes, Decision Tree, and Feedforward Neural Network (MLP) to classify animals from images with a total of 1,512 images are provided divided equally among three animal classes chicken, horse, and sheep with 504 images per class. All of the images have been resized to 32x32 pixels and flattened to create feature vectors.

The Naive Bayes classifier makes use of basic statistics (mean and variance of pixel intensities) as baseline features, while the Decision Tree classifier used the raw pixel intensities with visualizations created to provide interpreted decision paths. Finally, the Feedforward Neural Network used the MLPClassifier in Scikit-learn with one and two hidden layers to learn more complex relationships within the data.

The models are compared on their classification accuracy, precision, recall, F1-score, and confusion matrices so that the models can be understood on how best they differentiate among the three animal classes. The findings provide tradeoffs between model simplicity, model interpretability, and model predictive performance with image classification in a small data setting.

Contents

Table Of Figure

Introduction and Dataset Description:

One of the fundamental tasks in machine learning and computer vision is image classification which consists of classifying images into relevant classes automatically. This application involves classifying images of animals into three classes: chicken, horse, and sheep. Given the rapid advancements in deep learning models for image recognition and the increasing complexities in image recognition, it is necessary to explore many alternatives in classification methods.

The dataset contains 1,500 images, and each class has 500 images. Images in the dataset have been sourced and sorted into an organized structure; each class is housed in its own directory. The images in the dataset have been resized to 32x32 pixels to ensure all of the samples are uniform from the outset. Resizing the images help to extract features and allows machine learning algorithms to process images efficiently.

In this application, there will be three separate models: Naive Bayes, Decision Tree, and a Feedforward Neural Network (MLP). The Naive Bayes classifier uses basic statistics to take features derived from pixel intensities as a simple baseline model. The Decision Tree classifier uses raw pixel values, and it allows for visually interpretive decision paths. The MLP classifier that is explored in this section is implemented using Scikit-learn, and it captures complex relationships in images.

Detailed explanation of each model.

In this project, there are three classification models implemented for the purpose of classification of the animals' images being the Naive Bayes, Decision Tree, and Feedforward Neural Network (MLP). They have different advantages and differ in the way to approach the image data.

1. Naive Bayes Classifier

The Naive Bayes classifier is a machine learning model that is probabilistic and is grounded on Bayes' Theorem where we assume that given a class label, each input feature is conditionally independent of the others. In practice, this means that Naive Bayes models are most useful with small data sets but work very well even when the input space is high-dimensional, such as is often the case with any image classification task.

Basic elements:

- **Statistical Feature Extraction:**

To summarize the information in each image, for this implementation we extract the mean and the variance of the pixel intensities for each image. In this way, we can summarize all of the important information in the image with a compact representation.

- **Training Mechanism:**

Through the learning process, the model estimates the likelihood for each class from the statistical features, obtaining probability distributions for the data, across each class. These distributions are used in the representations to predict.

- **Evaluation Metrics:**

This classifier is evaluated using accuracy, precision, recall, the F1 score - all of which provide a set of balanced measures of the performance of the classifier.

2-Decision Tree Classifier

The Decision Tree classifier is a machine learning model that is non-parametric in nature. The Decision Tree classifier builds a tree structure by recursively splitting the dataset into smaller subsets of data by features. The decision tree's structure is tree-like in that, **Each node represents a decision based on some feature**, **Each branch represents an outcome or rule**, and **Each leaf represents a final class assignment**.

Basic elements:

- **Interpretability:**

One of the key benefits of a decision tree is that it is transparent. You can literally see each decision the model makes step-by-step and it does not take long to explain why that prediction was made great for debugging or trying to communicate the results with humans.

- **Raw Pixel Data:**

Unlike the Naive Bayes model, which uses summary statistics, the Decision Tree classifier deals directly with raw pixel intensity values, which allows it to capture more detailed and complex patterns within the image data.

- **Overfitting Control:**

With a tree, it is pretty easy to be too specific (overfit to the training data), so we can avoid this by restricting the tree's depth (max depth = 15, for example). This way we can help the model to generalize and have better predictions on new, unseen images.

3-Feedforward Neural Network (MLP)

The Feedforward Neural Network, or Multi-Layer Perceptron (MLP) is a type of artificial neural network made up of layer-by-layer connections of neurons. Each neuron acts on inputs through weighted sums and activation functions and data flows strictly forward from input to output.

Basic elements:

- **Hidden Layers:**

In this project, we use the MLP with one or two hidden layers. This allows for the MLP to learn deeper and more abstract relationships in the image data than simpler models.

- **Data Normalization & Standardization:**

The pixel values were normalized to [0, 1], then standardized (zero mean, unit variance). This improves both training speed and stability - improving learning and convergence.

- **Activation functions:**

Using non-linear activation functions such as ReLU, allow the MLP to learn complex non-linear patterns in the data - that linear models would completely miss.

Summary

The fusion of Naive Bayes, Decision Tree, and Feedforward Neural Network (MLP) offers a well-rounded and thorough approach to image classification because each model contains unique advantages: Naive Bayes has simplicity; Decision Trees provide transparency and interpretability; MLP's have the ability to use deep learning to identify complex patterns. In addition to side-by-side comparison, we can learn a lot about how each model works with image data for observable features and small differences in advantages. Ultimately, comparison of this nature teaches us a lot about the behavior of the model which will help later in this chapter when determining which model is the best for classification problems.

Results and Discussion

The dataset contained 1,500 images broken down evenly into three classes, chicken, horse, and sheep, with 400 training samples and 100 testing samples for each class. The models were tested on a test set of 300 samples using precision, recall, F1-score and accuracy together with confusion matrices.

```
Total samples:    1500
Training samples: 1200
Testing samples:  300

Samples per class:
      chiken: train= 400  test= 100
      hourse: train= 400  test= 100
       sheep: train= 400  test= 100
```

Figure 1:run code images distribution.

1-Naive Bayes Classifier

- **Performance:** The Naive Bayes classifier achieved an overall accuracy of **47%**.
- **Class-wise observations:**
  - The highest recall was for **chicken (72%)**, indicating relatively better detection for this class.
  - The recall for **sheep was notably low (15%)**, highlighting a significant challenge in classifying this class.
  - Precision was highest for **horse (60%)**, but overall performance was limited.
- **Confusion Matrix Insights:** The model struggled with distinguishing sheep from the other classes, as evidenced by the confusion matrix.
- **Summary:** This baseline model demonstrates some basic classification ability but is insufficient for robust differentiation among classes.

```
=== Naive Bayes Classifier (Mean & Variance Features) ===
              precision    recall  f1-score   support

      chiken       0.43      0.72      0.54       100
      hourse       0.60      0.55      0.57       100
       sheep       0.38      0.15      0.21       100

    accuracy                           0.47       300
   macro avg       0.47      0.47      0.44       300
weighted avg       0.47      0.47      0.44       300

Confusion Matrix:
[[72 14 14]
 [34 55 11]
 [62 23 15]]
```

Figure 2:Naive Bayes report.

2- Decision Tree Classifier

- **Performance:** The Decision Tree classifier achieved an improved overall accuracy of **59%**.
- **Class-wise observations:**
  - Precision and recall were relatively balanced across classes, with values around **56-62%**.
  - Sheep classification improved compared to the Naive Bayes model, with recall at **61%**.
- **Confusion Matrix Insights:** The Decision Tree showed better class distinction but still faced challenges, particularly with horse and sheep classifications.
- **Interpretability:** The visual representation of the decision tree aids in understanding the classification process.
- **Summary:** This model effectively utilizes raw pixel intensities and decision rules, providing a moderate improvement in classification accuracy.

```
=== Decision Tree Classifier ===
              precision    recall  f1-score   support

      chiken       0.59      0.61      0.60       100
      hourse       0.62      0.55      0.58       100
       sheep       0.56      0.61      0.59       100

    accuracy                           0.59       300
   macro avg       0.59      0.59      0.59       300
weighted avg       0.59      0.59      0.59       300

Confusion Matrix:
[[61 13 26]
 [24 55 21]
 [18 21 61]]
```

Figure 3:Decision Tree report.

The decision tree visualization

The figure presents a visualization of the Decision Tree classifier's first three levels, illustrating how it classifies images of chickens, horses, and sheep based on pixel intensity values. Each node represents a decision point, where a specific pixel value is evaluated to guide the classification, with "True" and "False" paths leading to further splits. The leaf nodes indicate the predicted class, color-coded as green for chicken, orange for horse, and purple for sheep, along with sample counts at each node. This structure reveals the model's decision-making process and highlights which pixel features are most influential in distinguishing between the animal classes.

.

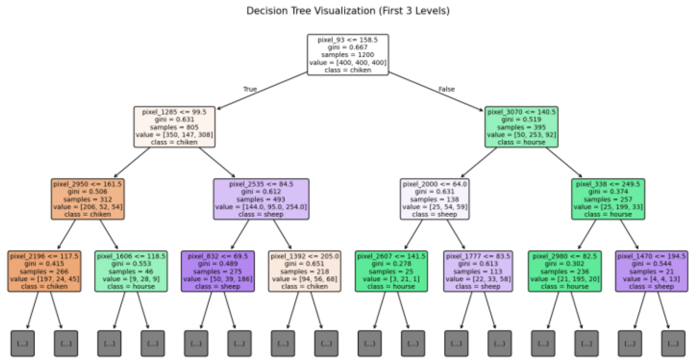

Figure 4:The decision tree visualization.

3-MLP Classifier

MLP Classifier (1 Hidden Layer):

- **Performance:** The feedforward neural network with one hidden layer achieved the highest accuracy of **79%**.
- **Class-wise observations:**
    - Strong precision and recall across all classes (precision 73-85%, recall 75-83%).
    - The model demonstrates effective learning of complex patterns, leading to robust classification.
- **Confusion Matrix Insights:** Misclassifications were minimal, particularly for chicken and horse, with sheep still showing some confusion.
- **Summary:** The MLP model benefits from learning non-linear features, significantly outperforming the simpler models.

```
=== MLP Classifier: 1 hidden layer (100 neurons) ===
              precision    recall  f1-score   support

      chiken       0.79      0.83      0.81       100
      hourse       0.85      0.78      0.81       100
       sheep       0.73      0.75      0.74       100

    accuracy                           0.79       300
   macro avg       0.79      0.79      0.79       300
weighted avg       0.79      0.79      0.79       300

Confusion Matrix:
[[83  5 12]
 [ 6 78 16]
 [16  9 75]]
```

Figure 5:MLP Classifier report (1 Hidden Layer).

MLP Classifier (2 Hidden Layers)

- **Performance:** The MLP with two hidden layers achieved slightly lower accuracy at **78%** compared to the single hidden layer model.
- **Class-wise observations:**
    - Precision and recall remained relatively stable but showed a slight decrease compared to the single hidden layer.
    - This could indicate potential overfitting or challenges in leveraging the additional complexity.
- **Summary:** While deeper architectures can capture more intricate patterns, careful tuning is necessary to optimize performance on this dataset.

```
=== MLP Classifier: 2 hidden layers (100, 50 neurons) ===
              precision    recall  f1-score   support

      chiken       0.79      0.81      0.80       100
      hourse       0.83      0.75      0.79       100
       sheep       0.72      0.77      0.74       100

    accuracy                           0.78       300
   macro avg       0.78      0.78      0.78       300
weighted avg       0.78      0.78      0.78       300

Confusion Matrix:
[[81  7 12]
 [ 7 75 18]
 [15  8 77]]
```

Figure 6:MLP Classifier report (2 Hidden Layer).

Comparative Analysis and Discussion of Model Performance

The comparison of Naive Bayes (NB), Decision Tree (DT), and Feedforward Neural Network (MLP) models reveals a clear pattern of performance and complexity:

- **Naive Bayes (NB)**: This model relies solely on the mean and variance of features, resulting in the lowest accuracy (47%). Its simplistic representation fails to capture spatial characteristics, particularly struggling to classify similar classes like sheep.
- **Decision Tree (DT)**: The DT model achieved improved performance (59%) by utilizing raw pixel data and providing visual decision paths. However, its reliance on discrete, axis-aligned splits can lead to overfitting, limiting its effectiveness.
- **One-Layer (MLP)**: The MLP with one hidden layer demonstrated the highest accuracy (79%). The hidden layer enables the MLP to learn complex, nonlinear structures, improving class generalization. Additionally, preprocessing steps like normalization and min-max scaling contribute to its strong performance.
- **Two-Layer MLP**: Although it showed comparable accuracy, the two-layer MLP faced minor issues with overfitting due to limited data, resulting in diminishing returns.

Overall, this comparison highlights that deeper and more expressive models, such as the MLP, can significantly outperform simpler models in handling complex data.

Conclusion

This project presented an evaluation of three classification models, Naive Bayes, Decision Tree, and an MLP, for classifying images of chickens, horses, and sheep, and in this case study, improved classification accuracy was a function of model complexity and capability to learn non-linear relationships in the data. Of the three models, why did Naive Bayes perform the worst? Naive Bayes was focused on the simple statistical features in the dataset - chicken, horse, and sheep images - and needed to account for the associated complexity in each of these objects. Decision Tree improved accuracy and interpretability of the model by using the raw pixel values as features instead of generated basic statistical features or summary statistics, however, this model relied on configuration with axis-aligned splits and was also using the raw pixel values which could result in overfitting the training data to an extreme level. The two-layer MLP with one hidden layer was the best model based on best classification performance based on class label accuracy, class label balance, and ensemble agreement, as this model was able to learn complex / nonlinear relationships in the training dataset. The two-layer MLP performed less effectively than the one hidden layer model, and overfitting could be to blame - the underlying training dataset limited the effectiveness of the model. Based on the findings of this study, the MLP was the most effective classification model demonstrated in this study. To further improve improved classification accuracy and robustness, the potential to increase the training data set, use data augmentation, and use advanced models like CNN, it is recommended.